DATA ITEM DESCRIPTION

Title: ACCEPTANCE TEST EQUIPMENT (ATE) VALIDATION PLAN

Number: DI-MISC-80759B
AMSC Number: N6735
DTIC Applicable: No
Office of Primary Responsibility: MI
Applicable Forms: N/A

Approval Date: 20220407
Limitation: N/A
GIDEP Applicable: No
Project Number: MISC-2021-013

Use/Relationship: The ATE Validation Plan provides the philosophy, test logic, sequence and test schedules, and test procedures to be utilized in the development, acceptance, and control of the Acceptance Test Equipment (ATE).

a. This Data Item Description (DID) contains the format and content preparation instructions for the data product generated by the specific and discrete task requirement as delineated in the contract.
b. This DID is applicable to acquisitions of military systems, equipment, and components that require the use of acceptance test equipment to achieve the acceptance of Military items and the traceability to that items technical data requirements.
c. When this DID is imposed in a contract or purchase order, the procuring activity must ensure that DI-SESS-81004, Special Inspection Equipment (SIE) Engineering Design Data and Associated Lists is applied at the appropriate level. When ATE is used by a Depot, then use full application of the DID; otherwise use as a guide.
d. This DID supersedes DI-MISC-80759A, DI-QCIC-80757A, and DI-QCIC-80922A.

Requirements:

1. Content: The ATE Validation Plan shall include the ATE Concept, Pre-Validation Activities, Validation Procedure, and Validation Certification. The ATE Validation Plan shall also include a Compliance Matrix that traces the ATE capabilities to test the product acceptance requirements, and a proposed Fault List as Appendices.
2. Format: The ATE Validation Plan format shall be in accordance with the following Validation Plan sample:

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The Validation Plan presents the ATE Concept, Pre-Validation activities, Validation Procedures, and Validation Certificate activities to be utilized in the development of ATE.

I. ATE CONCEPT

The Acceptance Test Equipment Concept details design performance calibration and accuracy requirements information to be used for screening and approval of recommended ATE prior to design and fabrication of the ATE. It is used in design control to screen each proposed design against the procuring activities' existing inventory of equipment to satisfy the requirement; maintain management control over the efforts of the contractor in the area of inspection equipment; and assure integration of design of acceptance inspection equipment. The ATE Concept format shall be contractor selected and include the following:

A. Physical Description

B. Functional Description

C. Equipment Options (including costs)

D. Concept sketch of equipment

E. Block diagrams

F. Requirements for calibration equipment external to the ATE

II. PRE-VALIDATION ACTIVITIES

A. **Theory of Operation** - The Contractor shall provide a summary of how the ATE is intended to be used and how it operates to perform its intended function before validation.

B. **Test Equipment HW/SW and Procedures to be Validated** - The Contractor shall provide in advance of validation the list of all test sets, software, drawings, ATP/Calibration procedures and any other document that will be validated and placed under configuration control.

C. **Test Schedule and Contractor Cognizant Employees** - The contractor shall provide a schedule of events/activities for the government validation procedure and a list of cognizant employees for each activity.

D. **Internal Verification (Prove-in)** - The Contractor shall conduct an internal verification before government validation to ensure that the test equipment to be validated demonstrates that it performs to its intended function. Data collected during this activity will be reviewed and may be used as part of the repeatability analysis.

E. **Acceptance Limits for Performance Capabilities** - The Contractor shall provide a list of the applicable documents and drawings that contain the testing requirements including the contract number and MIS number. The Contractor shall trace the MIS SPEC requirements against the test set testing capabilities. The Contactor shall generate a requirement vs. SW modules compliance matrix (See Appendix A as an example). This will be performed by tracing software test modules to each of the requirements in the applicable system or

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component performance specification and the applicable drawings, if any. (A test module is considered to be the source code for a given test requirement). The matrix can be generated in advance and verified at validation.

III. **VALIDATION PROCEDURE**

Note: This section could be used as a Validation Checklist.

A. **Entry meeting activities**:

1. Review the test station's theory of operation.

2. Verify that the test set has undergone an internal verification and it has been accepted by the proper department/organization.

3. Review data collected during test station's internal verification or prove-in.

4. Check on status of "Golden Units" (review data). (A Golden Unit is defined as a unit under test (UUT) that has been certified as a good unit by an approved or validated acceptance test set).

B. **Software activities**:

1. Verify that software documentation has been properly signed off (internal verification).

2. Verify that the software is under configuration control IAW established procedures.

3. Review and verify the requirement traceability/Compliance Matrix.

C. **Configuration Audit (HW/SW) Activities:**

1. Review the cable interconnects and overall station setup vs. the test set drawing package.

2. Perform schematic traces of all test requirements (i.e., how are the connections made to perform given test to a given requirement, to include the software module that requests the test). Depending on the number of tests, this could be performed on a sample basis.

D. **Calibration Activities:**

1. Review and verify the test set calibration procedures (if any).

2. Verify test set overall calibration sticker is in place.

3. Verify that individual instruments are calibrated and clearly show their corresponding calibration stickers.

4. Review test equipment test error analysis.

5. Review contractor's calibration recall system.

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E. **Test Set Activities:**

1. Perform a software up-load from the software configuration management libraries (i.e., web site, disc, CD, etc.).

2. Verify that the test procedure (TP)/operating instructions (OI) have been signed off by the proper departments/organizations.

3. Baseline runs: Perform a baseline run by running a complete functional test with a known golden unit by following all the steps in the TP/OI. If a golden unit is not available, proper arrangements should be made for an appropriate substitute before on-site validation. The paragraphs on the TP/OI will be used for test cross-reference and should be added to the Compliance Matrix (Appendix A).

4. Fault insertion: This step will verify that the test set(s) can identify faults in the UUT. The Contractor shall provide up to ten non-destructive faults before the on-site validation. The Government representative has the option of using all of the contractor's generated faults or select their own. Each fault insertion will be followed by a "good run" (only for that particular test). The data from the good runs can be used later for the repeatability step. These runs will be conducted at ambient temperature.

5. Repeatability: After all fault insertion has been performed, another baseline run will be performed at the option of the Government representative. This data, along with the previous gathered data, will be used to determine the repeatability of the test set(s).

F. **Configuration Control:**

1. At the completion of Government validation, the test set documentation (HW/SW) shall be under the same configuration control as the product documentation.

2. At the request of the Government representative, the Contractor shall explain their configuration control procedures as they apply to the test set's software and hardware IAW their written procedures.

G. **Validation Process/Results:**

1. The Government representative and Contractor personnel should have a meeting at the end of each day of validation to discuss how the validation is proceeding. Any problems that occurred that day should be discussed to determine impact to the validation. If no problems were discovered, the attending members should discuss the next day's agenda.

2. Once all validation activities have been completed, ensure that the test station is placed under strict configuration control and tamper proof seals are placed on the validated test set.

3. As part of the validation process, the following shall be considered:

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a. Test set maintenance spares.

b. Establish maintenance records and schedules for the test set.

c. Ensure all maintenance performed on the station shall be recorded as performed.

4. Unless contractor has a procedure in place that follows all test set activities, a logbook that shows all the above work shall be generated.

IV. **ATE VALIDATION CERTIFICATION**

A. **Validation Data -** The Government representative will compile all validation data for review and approval. The Government validation is deemed complete when the following is satisfied:

1. All test equipment documentation has been properly signed off by the Contractor's quality engineering activity and placed under configuration control.

2. The test equipment has demonstrated that it does perform to its intended function.

3. A compliance matrix has been generated and covers all the acceptance requirements of the UUT.

4. All action items generated during validation are resolved and closed or a path forward has been identified.

B. **ATE Validation Certificate:**

1. Once all test equipment validation requirements are satisfied, a validation certificate attesting to the successful validation of the test equipment will be issued. The validation certificate should be signed by all personnel involved in the validation activities.

2. The validation certificate will be valid until such time that changes are made to the test set HW/SW and/or procedures. At such time, the Government reserves the right to re-validate the test set. The validation certificate will remain with the corresponding test set as proof that the test equipment is adequate for hardware acceptance.

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APPENDIX A
Compliance Matrix

This matrix is made-up from the specification acceptance requirements and any other column that reference the validation crosswalk between SW, HW, documentation and requirements. In this example the first three columns are taken from the MIS spec. The fourth column would be the reference for each SW module that perform the test; the fifth column is taken from the test set operating procedures and the last one is to record any events that pertain to that test. It should be noted that only QCI or acceptance requirements should be part of this matrix.

Description	**MIS Specification Paragraph**	**Mode of Test in MIS QCI**	**SW Module Name**	**ATP Compliance Paragraph**	**Comments Special circumstances, events and notes**
Components	3.2.1	E	Comp.txt	2.2.1	ECP MI-M2120 removes AT verification
Azimuth displacement	3.4.1.1.1	D	AZ.cll	2.2.1	ECP MI-M2120 removes AT verification
Elevation displacement	3.4.1.1.2	D	Eldis.dll and eldispl.txt	2.2.2	ECP MI-M2120 removes AT verification
TU Locking	3.4.1.2	D		5.2.2	ATP 5.2.2
Tracking rate	3.4.1.3	T		5.2.5	ATP 5.2.5
Damping rates and breakout torque	3.4.1.4	T		5.2.6,	ATP 5.2.6 breakout temp requirement from ambient, make temp test fat only
Equilibrator	3.4.1.5	T		5.2.3.5	ATP 5.2.3.5
Input power	3.4.1.7.1	T		5.2.7	ATP 5.2.7
Holding torque	3.4.1.7.2	T		5.2.8	ATP 5.2.8
Left handgrip controls	3.4.1.8.1.1	T		5.4.5	ATP 5.4.5
Rotary release mechanism	3.4.1.8.1.2	T			ECP MI-M2120 removes from table IX
Brake switch	3.4.1.8.1.2.1	T	T	Y	ATP 5.4.5
Adjustment travel	3.4.1.8.1.2.2	T	T	Y	ATP 5.2.11

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APPENDIX B
Fault Insertion Check List
(Check the ones performed and annotate their corresponding results)

1. UUT1 interface fault#1: Remove TAS cable connection prior to electrical test.
   Result: All circuits listing connector 2J2 as termination point will fail.

2. UUT1 interface fault#2: Remove COIL cable connection prior to electrical test.
   Result: All circuits listing connector 2W1P2 as termination point will fail.

3. UUT1 interface fault#3: Lower Arming Lever prior to electrical test.
   Result: All circuits listing connector 2P1 as termination point will fail.

4. UUT1 interface fault#4: Do not press any Left handgrip switch during electrical test.
   Result: All circuits listing connector 2J2-58 as termination point will fail.

5. UUT1 interface fault#5: Do not press any Right handgrip switch during electrical test.
   Result: All circuits listing connector 2J2-69 as termination point will fail, as well as TOW missile circuit between 2W1P2-124 and 2W1P2-125.

6. UUT2 interface fault#1: Remove TAS cable connection prior to electrical test.
   Result: All circuits listing connector 2J2 as termination point will fail.

7. UUT2 interface fault#2: Remove COIL cable connection prior to electrical test.
   Result: All circuits listing connector 2W1P2 as termination point will fail.

8. UUT2 interface fault#3: Lower Arming Lever prior to electrical test.
   Result: All circuits listing connector 2P1 as termination point will fail.

9. UUT2 interface fault#4: Do not press any Left handgrip switch during electrical test.
   Result: All circuits listing connector 2J2-58 as termination point will fail.

10. UUT2 interface fault#5: Do not press any Right handgrip switch during electrical test.
    Result: All circuits listing connector 2J2-69 as termination point will fail, as well as TOW missile circuit between 2W1P2-124 and 2W1P2-125.

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